

	Best case	Average case	Worst case
Algorithm 1	$O(n)$	$O(n)$	$O(n)$
Algorithm 2	$O(1)$	$O(n)$	$O(n)$

What is the lower bound of the problem?

Lower bound of the problem is $\Omega(n)$ since we must check all the elements to confirm they are valid that they don't belong more than one category, they are not supposed to overlap.